Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Thread B · Day 1 — Angle Pairs: Supplementary, Complementary, Vertical, Adjacent	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.4

2. Standards

MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

Must. Name adjacent, vertical, complementary and supplementary pairs (a linear pair is supplementary); find a missing angle from each; write and solve the equation when the angles are expressions, and report the angle the question asks for.

Must not. Parallel lines and transversals (Geometry); angle sums of polygons (MA.8.GR.1.6); reflex angles.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II–III: one pair as a picture, a name and an equation
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 7 split; the re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes III and boards 5–6: the same equation shape behind every pair

4. Learning Target and Essential Question

Learning target. I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

Essential question. When two angles are neighbours or opposites, what does one tell you about the other?

Building on. Grade 4 (MA.4.GR.1.3) — angle measures add, a straight angle is 180°; Unit 1 — equations with the variable on both sides.

Working toward. T-B2 tomorrow — the angles of a triangle; Geometry — angle pairs made by parallel lines and a transversal.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Can two adjacent angles be vertical angles? Why or why not?	DOK 2
Two angles are supplementary and equal. What are they, and what do they make?	DOK 2
Board 6's x is 12 and its angles are 52°. Which does the question ask for, and how do you know?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	What is the complement of 34°?	***56°***	146° — used 180 [Source of Truth, GR.1.4: complementary and supplementary swapped]
2	What is the supplement of 112°?	***68°***	−22 or 'none' — used 90 [Source of Truth, GR.1.4: complementary and supplementary swapped]
3	Two lines cross. One angle is 47°. Find a and b.	***a = 47°, b = 133°***	a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]
4	Angles P and Q are complementary, and angle P is 28°. What is angle Q?	***B — 62°***	(A) 180 − 28: supplementary for complementary [Source of Truth, GR.1.4: complementary and supplementary swapped]; (C) took the two as equal — that is vertical angles [Notes I]; (D) 90 + 28: added instead of subtracting [Notes II: the two add to 90]
5	A linear pair measures $(3x+15)^\circ$ and 105°. Find x.	***x = 20***	30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]; −5 — used 90 [Source of Truth, GR.1.4]
6	Vertical angles measure $(6x-20)^\circ$ and $(2x+28)^\circ$. Find the measure of each angle.	***52°***	12 — reported x as the angle [Math Nation names reporting x]; 22 — added them to 180 [vertical angles are equal, not supplementary]
7	In the figure, lines AD and BE cross at O, and ray OC is drawn. Which two angles are adjacent?	***C — angles AOB and BOC***	(A) a vertical pair: across the crossing, no shared side [Notes I]; (B) angle AOB lies inside angle AOC — the two overlap [Notes I: no overlap]; (D) these share only the vertex O, not a side [Notes I: a shared side]
8	Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°. Find x.	***x = 142***	92 — used the 50° (180 − 38 − 50) [the extra number]; 52 — used 90 [Source of Truth, GR.1.4]
9	Can two vertical angles be complementary? Always, sometimes or never? Explain.	***Sometimes — only when each is 45°.***	'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]; 'always' [Source of Truth, GR.1.4]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. One card on the desk: Corner (90) — complementary; Straight (180) — supplementary; across — vertical, equal; next to, sharing a side — adjacent. Name the pair before writing a number.

ELL. **Adjacent**: next to. **Vertical** here means across from (not up and down). **Complement / supplement**: the piece that completes 90 / 180. Point and say each one.

Enrichment. Three lines cross at one point and make six angles. Two of them are 40° and 75°. Find the other four. (40°, 75°, 65°, 65°: vertical pairs are equal, and the three around one side make 180.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: vertical angles are equal, so they are complementary only when each is 45° — sometimes, never always. A board that says 'yes' or 'no' without the 45 has the verdict and not the reason.

Vocabulary. **adjacent angles**; **vertical angles**; **complementary angles**; **supplementary angles**

Materials. Whiteboards and markers. Calculators allowed, printed nowhere. Wednesday is 43 minutes: the cut is boards 3 and 7 and two minutes of the IXL block, never board 9; the timing table totals 53 by house rule.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Identify complementary, supplementary, vertical, adjacent, and congruent angles — HGV; Find measures of complementary, supplementary, vertical, and adjacent angles — R2B; Write and solve equations using angle relationships — DBV.